When Routing Benchmarks Reward Repetition:
Evaluating Recipient Selection in a Persistent Agent Mesh

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Abstract

Routing a request to one of several persistent agents is a decision that is repeated by the same sender many times, and most repetitions select the agent the sender addressed last. An evaluation that scores all decisions alike therefore measures, in large part, whether a policy reproduces the sender's previous choice. We formalise a history-conditioned evaluation protocol for recipient selection and apply it to 908 routing decisions reconstructed from the event journal of a production agent mesh. On 555 worker-directed decisions, a small state-informed classifier agrees with the recorded recipient on 57.8%, a recent-contact rule on 60.0%, and a time-ordered per-sender majority on 56.0%. Conditioning on history decomposes these totals into opposing components: relative to the majority rule the classifier gains 22 agreements on recipient switches, loses 31 on continuations, and gains 19 on decisions without exchange history. Under a linear reweighting of the observed conditional scores, the ranking of the two policies reverses at a continuation share of 0.58, below the observed share of 0.66. An activity mask, an analytic novelty control, a lexical audit and clustered inference bound the explanations of the residual signal. The same protocol applied to three frontier language models locates the ceiling of state-based selection: Claude Opus 5.5 reaches 0.775 overall and 0.568 on switches; with activity fields masked it retains 0.467 on switches but falls to 0.369 on continuations, below the majority rule. Across all models tested, agreement on continuations is attributable to recency encoded in the state, whereas agreement on switches is determined by the quality of the reader. We release the reconstruction procedure, the replay harness and the metric, and register a frozen arrival-triggered override policy for prospective validation.

1 Introduction

Systems in which language-model agents exchange messages are now common: an agent assigned to one repository requests a review from an agent assigned to another, an orchestrating agent dispatches work to the peer that already holds the relevant working tree, and a human relays instructions through a messaging bridge. In such a system the recipient of a request is a long-lived process with a current task, a history with the sender, and a finite lifetime. It is not a model drawn from a pool. Senders currently name recipients explicitly. A natural extension is to have a model select the recipient, and the natural evaluation is the rate at which the model's selection agrees with the recipient the sender would have named.

This paper concerns the semantics of that rate. In the mesh studied here, two of every three requests are addressed to the peer with whom the sender most recently exchanged messages. A policy that observes only the sender's previous recipient is correct on all of those decisions and incorrect on all others, by construction. A learned policy evaluated on the pooled stream is thus scored predominantly on decisions that a trivial rule already resolves. More consequentially, we

show that the pooled score can rank two fixed policies in either order as a function of a quantity neither policy controls: the proportion of repeated contacts in the evaluation stream.

The contributions are as follows.

1. **A history-conditioned evaluation protocol** (Section 3). Each decision is assigned, from information available before the request, to one of three history conditions: *continuation*, *switch*, or *no history*. Agreement is reported per condition beside the analytic chance level of the decision's candidate set, against temporal baselines that use no future information, with paired tests and an explicit dependence unit.

2. **An empirical counterexample** (Sections 4.1–4.2). From 908 decisions reconstructed from a production mesh's event journal, we show that the aggregate ranking of a learned policy against a majority rule is determined by the request mix; that masking activity-derived fields attributes the continuation score of every model tested to recency; and that the residual on switches survives a lexical audit but not every clustering of the paired test.

3. **The ceiling of state-based selection** (Section 4.3). Three frontier models evaluated under the same protocol establish that the limiting factor on switches is the quality of the reader of the state rather than the information content of the state, and that even the strongest model is dominated by the recent-contact rule when continuations exceed 0.88 of history-bearing traffic.

4. **A registered design consequence** (Section 4.4). The decomposition locates the useful work of a learned policy on switches and on decisions without history. We derive an override policy triggered by peer arrival, fix its parameters, and register a prospective test.

Throughout, the measured quantity is termed *recipient agreement*: the rate at which a policy's selection coincides with the recipient recorded in the sender's request. This label is reproducible and behaviourally grounded, but it is not evidence that the sender's choice was optimal, and no result here concerns downstream task outcome.

2 Problem setting

2.1 Persistent agents and attested state

The system under study is a daemon that registers agent peers and delivers addressed messages among them. Peers are coding agents running on several backends, a human relay, and job runners. Each peer carries *claimed* state, a free-text self-description that the peer sets and that decays in accuracy, and *attested* state that the daemon observes directly: backend, project (parsed from the peer name), liveness and the age of the last observed activity, turn activity, the count of open inbound requests, and the time of the peer's last exchange with the sender. Figure 1 depicts a single decision. The period studied, 22 May to 10 August 2026, predates the recording of routing snapshots, so descriptions are unavailable and every result concerns attested state alone.

2.2 Decisions, policies and agreement

Definition 1 (Routing decision)**.** A routing decision is a tuple $d = (s, x, \mathcal{C}, y, t)$: a sender s, a request text x, a candidate set $\mathcal{C}$ of peers with their attested state as of time t, and the recorded recipient $y \in \mathcal{C}$. The candidate set comprises every peer the daemon would have offered at t, excluding s and machine identities.

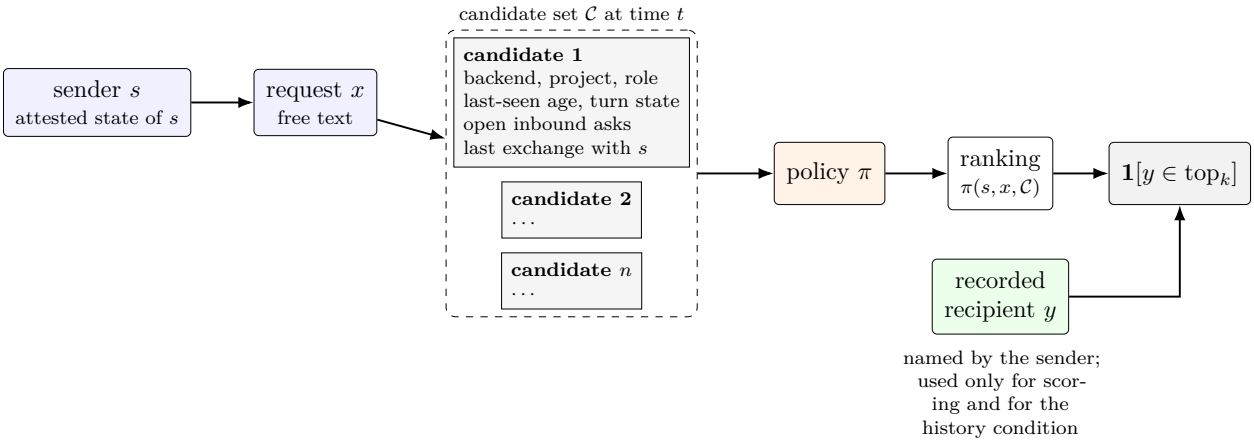


Figure 1: A routing decision. The policy observes the sender, the request text and the candidate set with attested state as of the request time, and returns a ranking. The recorded recipient is the label.

Definition 2 (Policy and agreement)**.** A policy π maps $(s, x, \mathcal{C})$ to a ranking of $\mathcal{C}$, or abstains. For a set of decisions $\mathcal{D}$ and rank k, the agreement of π is

$$A_k(\pi; \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \mathbf{1}\big[y_d \in \text{top}_k\, \pi(s_d, x_d, \mathcal{C}_d)\big], \tag{1}$$

where an abstention or failure contributes zero. We report $k \in \{1, 3\}$ and write A_1 when k is clear.

Definition 3 (Chance)**.** The chance level of $\mathcal{D}$ at rank k is the expected agreement of a uniform draw, $\bar{c}_k(\mathcal{D}) = |\mathcal{D}|^{-1} \sum_d \min(k, |\mathcal{C}_d|)/|\mathcal{C}_d|$. It is computed analytically rather than sampled.

2.3 History conditions

Let r_d denote the peer with whom s_d most recently exchanged messages before t_d, where an exchange is a request counted at its creation or at its first reply carrying a message, and expiring 24 hours after creation. Figure 2 illustrates the three cases.

Definition 4 (History condition)**.** A decision d is a *continuation* if r_d exists and $y_d = r_d$; a *switch* if r_d exists and $y_d \neq r_d$; and has *no history* if r_d does not exist.

The condition depends on the label y_d and is therefore a diagnostic partition for evaluation, not a feature observable by a policy at decision time. Section 4.4 addresses the distinct question of which observable signals predict membership in the switch condition.

2.4 Mixture dependence of the aggregate

For a policy π with conditional agreement $A^C(\pi)$ on continuations and $A^S(\pi)$ on switches, the agreement on a history-bearing stream with continuation share p is

$$A(\pi; p) = p\, A^C(\pi) + (1 - p)\, A^S(\pi). \tag{2}$$

Proposition 1. If $A^C(\pi_1) > A^C(\pi_2)$ and $A^S(\pi_1) < A^S(\pi_2)$, then there exists $p^* \in (0, 1)$ such that $A(\pi_1; p) > A(\pi_2; p)$ if and only if $p > p^*$, with $p^* = (A^S(\pi_2) - A^S(\pi_1))/[(A^S(\pi_2) - A^S(\pi_1)) + (A^C(\pi_1) - A^C(\pi_2))]$.

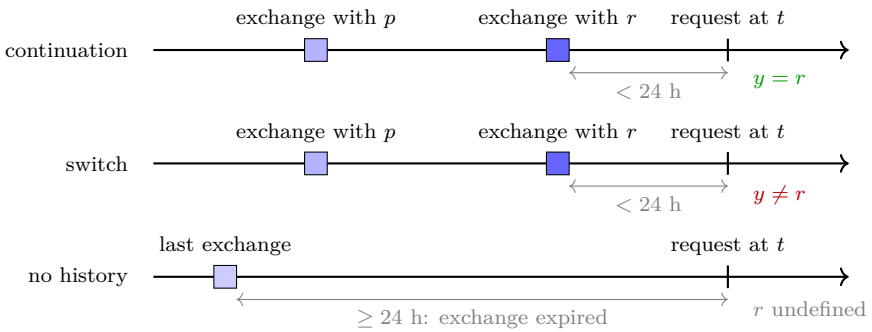


Figure 2: History conditions. r is the sender's most recent exchange partner within the 24-hour expiry; the condition compares the recorded recipient y with r. The condition is computed from the label and is used for evaluation only.

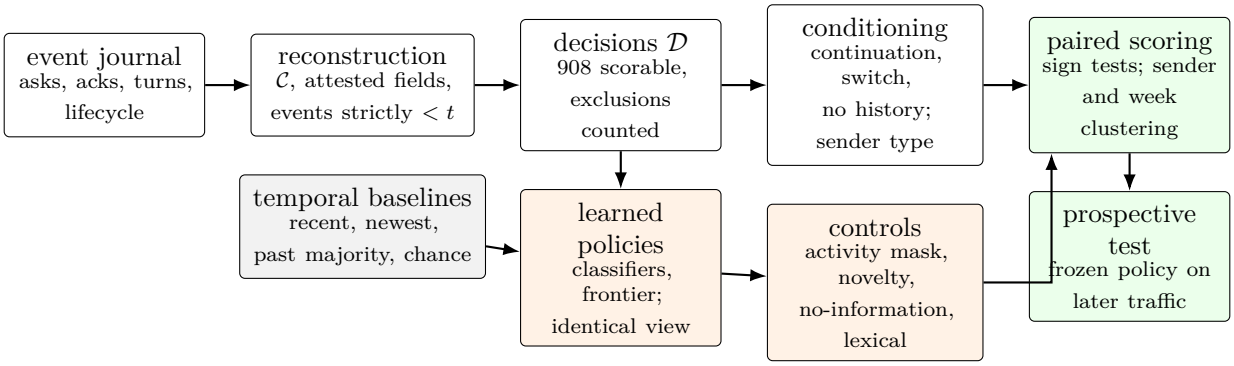


Figure 3: The evaluation protocol. Reconstruction uses only events before each request; conditioning uses the label; every policy receives the same view; controls test competing explanations; inference is paired and clustered; a derived policy is frozen before prospective validation.

The proposition is elementary. Its relevance is empirical: Section 4.1 shows that the premise holds for every learned policy tested against the temporal baselines, with p^* inside the range of continuation shares observed across routing datasets.

3 Protocol

Figure 3 summarises the protocol. Each stage is designed to exclude a specific alternative explanation of an apparent improvement.

Reconstruction. The daemon's event journal records requests (sender, recipient identifier, text, timestamp), acknowledgements, chat turns and peer lifecycle events. For each request we rebuild the candidate set and the attested fields from events strictly preceding t. Peer registration events begin only after the last request of the period, so liveness is inferred from activity within a 24-hour window; candidate sets are consequently a lower bound on the registry, and the window is a reported free parameter. Turn state is the type of the most recent chat turn within five minutes. Open inbound requests are those not closed by an acknowledgement or reply. Exchange semantics replicate the live daemon's tracker; four declared divergences and their per-row effect are given in Appendix A.

Table 1: Characteristics of the evaluated population. The DecisionBench column is a contrast in history structure (Section 4.5), not a replication.

Property	Historical mesh	DecisionBench aware-c2
Evaluated decisions	555 worker-directed, of 908 scorable	691 delegations
History conditions	333 continuation, 169 switch, 53 no history	93 same, 270 switch, 328 first in task
Repeat share, with history	$333/502 = 0.663$	$93/363 = 0.256$
Repeat share, all rows	$333/555 = 0.600$	$93/691 = 0.135$
Label	Recorded recipient of the request	Recorded delegate of the orchestrator
History semantics	Reconstructed exchanges with expiry	Previous delegation within task
Candidate information	Activity-inferred registry, attested fields	Static model cards

Population. Of 2,545 requests, 1,505 originated from machine dispatch and 39 were self- or service-addressed; both classes are excluded. Of the remaining 1,001, the recorded recipient was not observably online for 93 (75 with no prior evidence of existence, consistent with peers addressed at the moment of spawning). The 908 scorable decisions comprise 353 addressed to an orchestrator, which admit a single answer and are excluded from the routing analysis, and 555 addressed to a worker. Table 1 characterises the latter. Candidate sets have median 12 and mean 16.5 members (maximum 110). The sender is an orchestrator in 97 decisions, a peer in 243, and the human relay in 215.

Temporal baselines. *Recent* selects r_d and abstains when none exists. *Newest* selects the candidate with the smallest last-seen age. *Past majority* selects the candidate that s_d addressed most often in decisions strictly before d, abstaining without a vote. *Retrospective majority* votes over the entire window including later decisions; it is reported solely as a leakage reference and is not a deployable policy. *Novelty* selects uniformly among candidates that appear in no prior request, scored by expectation.

Learned policies. Every learned policy receives the same byte-identical attested view of the sender, the request text and each candidate, and returns a ranking. The policies are: a small hosted classifier (Jev, TypeSafe), the instrument of the original study; a second small hosted ranker (Clef Flash, Cloudflare Workers AI); and three frontier models, Claude Haiku 5.5, Sonnet 5.5 and Opus 5.5, prompted with the same view under a structured output schema requiring one score per candidate (Appendix C).

Controls. The *no-information* control supplies the classifier with empty description fields only. The *activity mask* removes every activity-derived field (last exchange with sender, open inbound requests, last-seen age, turn state) from the sender block and from every candidate, leaving name, project, backend and role. The *lexical audit* partitions decisions by whether the request text contains a literal recipient address or a project mention (Appendix B).

Inference. Policies are compared on identical decisions with exact two-sided sign tests on discordant pairs. Because decisions share senders and calendar weeks, the principal tests are repeated under sender and week clustering with the number of blocks stated (Section 4.2). All thresholds examined on this data are reported as exploratory.

Table 2: Agreement counts at rank one by history condition, worker-directed decisions. Abstentions are scored as zero.

History condition	n	Recent	Past majority	Jev	Jev − majority	Opus 5.5
Continuation	333	333	268	237	−31	308
Switch	169	0	43	65	+22	96
No history	53	0	0	19	+19	26
All	555	333	311	321	+10	430

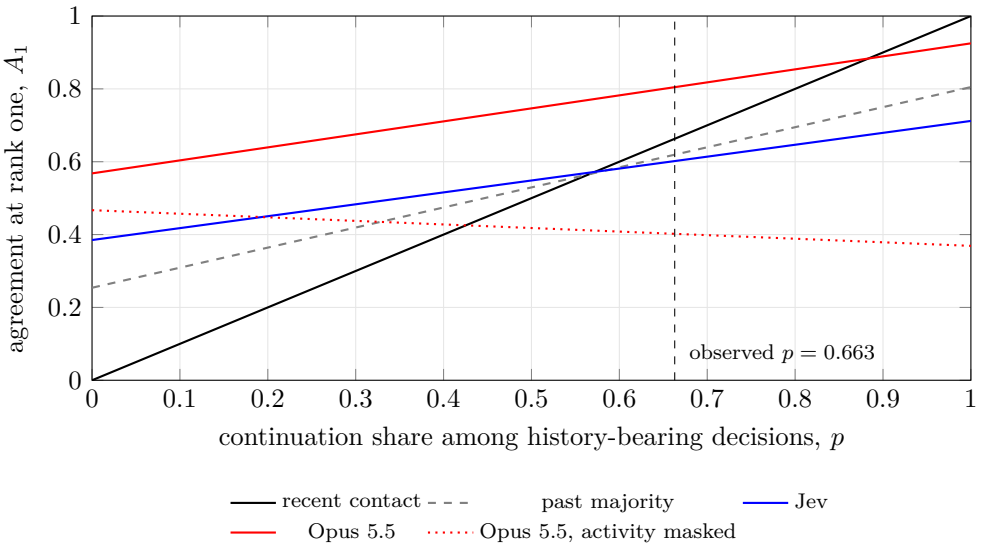


Figure 4: Aggregate agreement as a function of continuation share, equation 2 evaluated at the observed conditional scores of Table 2. The construction is a descriptive reweighting of recorded scores; no decision is re-executed, and decisions without history are excluded from the mixture.

4 Results

4.1 Decomposition of the aggregate

Table 2 reports agreement counts at rank one. Over the 555 worker-directed decisions, the classifier (321) is within one agreement of the recent-contact rule (333) and ten above past majority (311). Conditioned on history, the same decisions yield opposite-signed differences: on continuations the classifier trails past majority by 31 and recent contact by 96; on switches it leads past majority by 22 and recent contact by 65, the latter being zero by definition; on decisions without history it is the only one of the three that answers. The aggregate margin over past majority is accounted for entirely by the no-history condition. The switch margin and the aggregate margin are distinct effects.

Figure 4 plots equation 2 through these counts. For the classifier, $p^* = 0.583$ against past majority and 0.572 against recent contact; the observed history-bearing share is $p = 0.663$. The classifier is therefore dominated by both baselines on the observed stream and would dominate both on a stream with ten percentage points fewer continuations, with no change to any policy. The frontier models exhibit the same structure: Haiku and Sonnet cross the recent-contact rule at $p = 0.66$, coincident with the observed mix, and Opus 5.5, the strongest policy at both ends, is dominated by the one-line rule for $p > 0.883$.

Table 3: Agreement at rank one by history condition, worker-directed decisions. Chance is the analytic uniform expectation. Failures are scored as zero. Frontier and Clef rows are re-executions on the reconstructed snapshots (October 2026). The retrospective majority uses future votes and is a leakage reference only.

Policy	All 555	Contin. 333	Switch 169	No hist. 53	Orch. 97	Orch. sw. 62
Chance	.134	.170	.072	.109	.060	.061
Recent contact	.600	1.000	.000	.000	.289	.000
Newest	.571	.793	.195	.377	.258	.048
Past majority	.560	.805	.254	.000	.454	.339
Retrospective majority	.658	.808	.396	.547	.598	.500
No-information control	.139	.156	.095	.170	.113	.129
Clef Flash	.461	.556	.314	.340	.278	.290
Jev	.578	.712	.385	.358	.402	.371
Jev, activity masked	.312	.321	.302	.283	.196	.242
Haiku 5.5	.631	.757	.479	.321	.392	.387
Sonnet 5.5	.629	.766	.456	.321	.443	.403
Opus 5.5	**.775**	**.925**	**.568**	.491	.546	.500
Opus 5.5, activity masked	.389	.369	.467	.264	.340	.403

All policies by condition. Table 3 reports every policy on every condition. Three regularities hold. First, on continuations no learned policy exceeds the recent-contact rule, which attains 1.0 by construction, and only Opus 5.5 exceeds past majority. Second, on switches every learned policy exceeds the temporal baselines that can answer, and chance is an order of magnitude lower, because candidate sets are largest where the sender changes recipient (mean 23.3 candidates on switches against 12.3 on continuations). Third, the two small hosted rankers and the two smaller frontier models form a band: Clef Flash is below Jev on every condition, and Haiku and Sonnet exceed Jev on switches (25–9, $p = .009$; 28–16, $p = .10$) without exceeding the retrospective majority on any condition ($p \geq .14$).

Candidate-set size. Figure 5 plots agreement against the size of the candidate set. Chance falls from 0.455 to 0.027 across the four size classes, and every policy declines with it, but at different rates: past majority falls from 0.817 to 0.266, the classifier from 0.829 to 0.422, and Opus 5.5 from 0.951 to 0.523. The classifier exceeds past majority only in the largest class, where the latter's abstentions are concentrated. On switches alone, the ordering of policies is stable across classes. Pooled scores that omit the size distribution are therefore not comparable across datasets or conditions.

4.2 Explanatory controls

Agreement with the recorded recipient on switches may reflect a reading of the request against candidate state, or one of several lower-cost cues. Four are tested.

Activity recency. The activity mask removes every field derived from journal activity. For the classifier, continuation agreement falls from 0.712 to 0.321 and switch agreement from 0.385 to 0.302, the latter approximately four times chance and three times the no-information control. For Opus 5.5, continuation agreement falls from 0.925 to 0.369, below past majority, and switch agreement from 0.568 to 0.467. On continuations, therefore, the score of every model tested is attributable almost entirely to activity recency encoded in the state: both the classifier and the frontier model recover the sender's most recent exchange partner from the fields that encode it. On

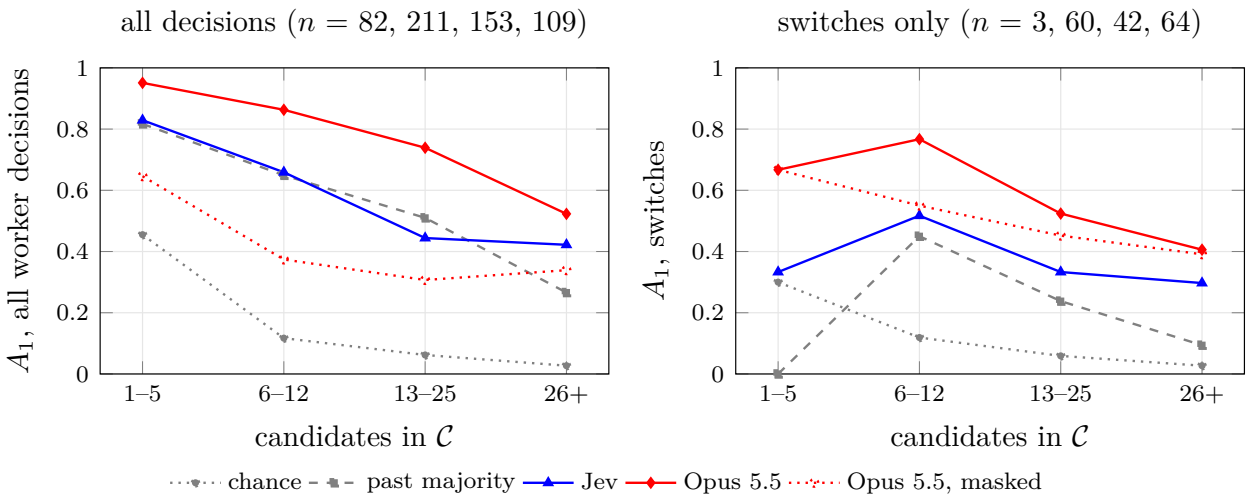


Figure 5: Agreement at rank one by candidate-set size. Chance varies by a factor of seventeen across size classes; policies decline at different rates. The smallest switch class contains three decisions and is shown for completeness only.

switches a residual persists that the mask does not remove, and whose magnitude depends on the reader (Section 4.3). Two qualifications apply. Candidate membership is itself activity-selected, so no field ablation removes activity entirely. The mask also removes the arrival signal available to a deployed policy, so masked scores understate deployable performance.

Novelty. Twenty-nine switches are addressed to a recipient absent from every prior request. On these, the novelty control attains an expected agreement of 0.478, exceeding the classifier (0.414), the masked classifier (0.414) and the newest rule (0.276). Conditional on the recipient being novel, a uniform rule over novel candidates is sufficient; the learned policy adds nothing measurable. Over all 169 switches the novelty control attains 0.082. Its value is therefore confined to knowing when to apply it.

Lexical cues. A literal recipient address (full display name, peer identifier or handle) occurs in 4 of 555 request texts, one of which is a switch that no policy resolves. Project mentions occur in 34% of switches. On the 111 switches without any detected recipient or project reference, the classifier attains 0.387 and the masked classifier 0.342, against 0.144 for past majority; on the 58 with a reference, past majority rises to 0.466 and the masked classifier falls to 0.224. The masked residual is thus not dependent on the request naming the recipient. This establishes robustness to a lexical filter, not semantic reading.

Dependence unit. The pooled switch margin of the classifier over past majority (46–24, $p = .012$) is retained under weekly blocking (8 blocks, $p = .031$) but not under sender blocking (52 senders, 15 positive against 7, $p = .13$; magnitude-weighted sign-flip $p = .29$). One sender contributes 16 of the 22 net additional agreements; excluding it yields 49 against 43 of 135 ($p = .50$). On the 62 switches issued by the orchestrator, which is the population routed by the live system, the classifier and past majority are indistinguishable (14–12, $p = .85$). The classifier's switch margin is therefore reported as nominal and concentrated in one sender's traffic. The frontier result is more robust under the same procedure: Opus 5.5 exceeds past majority on orchestrator switches by 17–7 ($p = .064$) and

Table 4: Paired comparisons at rank one on the 169 switches: wins–losses on discordant decisions with exact two-sided sign tests. The right-hand columns restrict to the 62 switches with an orchestrator sender.

Policy	Comparator	All switches		Orchestrator switches	
Jev	past majority	46–24	$p = .012$	14–12	$p = .85$
Jev, masked	past majority	45–37	$p = .44$		
Opus 5.5	retrospective majority	53–24	$p = .001$	15–15	$p = 1.0$
Opus 5.5	past majority	63–10	$p < 10^{-4}$	17–7	$p = .064$
Opus 5.5	Jev	38–7	$p < 10^{-4}$	13–5	$p = .096$
Opus 5.5	Sonnet 5.5	24–5	$p < .001$		
Opus 5.5	Opus 5.5, masked	27–10	$p = .008$		
Opus 5.5, masked	retrospective majority	51–39	$p = .25$		
Opus 5.5, masked	past majority	61–25	$p < .001$	17–13	$p = .58$
Opus 5.5, masked	Jev, masked	35–7	$p < 10^{-4}$		
Opus 5.5, masked	Jev	31–17	$p = .060$		

the classifier by 13–5 ($p = .096$), and exceeds both by wide margins pooled (Table 4).

4.3 The ceiling of state-based selection

The classifier is the instrument of the study rather than its object, and its switch residual admits two readings: a property of the state, or a property of the reader. Evaluating three frontier models on the identical view discriminates between them. Haiku and Sonnet behave as the classifier does, marginally above it on switches and below the majority baselines elsewhere. Opus 5.5 is the first policy to exceed the majority rule on every condition on which the comparison is defined: 0.775 overall against 0.658 for the retrospective majority (119–54, $p < 10^{-4}$), 0.925 on continuations (55–16), and 0.568 on switches (53–24, $p = .001$), from attested state alone.

The activity mask then separates the components of the frontier model's score. Continuation agreement falls to 0.369, below past majority, by the mechanism identified for the classifier. Switch agreement falls to 0.467, which remains level with the retrospective majority (51–39, $p = .25$), above past majority (61–25), above the masked classifier (35–7, $p < 10^{-4}$) and marginally above the unmasked classifier (31–17, $p = .06$). Figure 6 presents the decomposition. On switches, agreement separates approximately into 0.30 obtainable by any reader from the static state (project, role, backend, name), a further 0.17 attributable to the quality of the reading, and approximately 0.10 attributable to the activity fields. The static state therefore carries switch-relevant information that the small classifier does not extract and the frontier model does.

Cost. At list prices the 555 decisions cost \$0.28 with Haiku, \$3.67 with Sonnet and \$7.37 with Opus: approximately 1.3 cents per decision for the strongest policy, against a few hundredths of a cent for the classifier and no marginal cost for the temporal baselines. The frontier policies failed on 7 decisions each owing to a structured-output grammar limit at 79 or more candidates; the hosted ranker failed on 11 at 66 or more owing to an input cap. Both failure modes bias the corresponding columns toward smaller candidate sets.

Interpretation. The recent-contact rule remains the appropriate default on this stream: it exceeds every policy but one at no cost, and the exception costs two orders of magnitude more per decision. The ceiling of state-based selection on switches is, however, at least 0.57 rather than the classifier's 0.39, and approximately half of the difference persists after every activity cue is removed. Once a

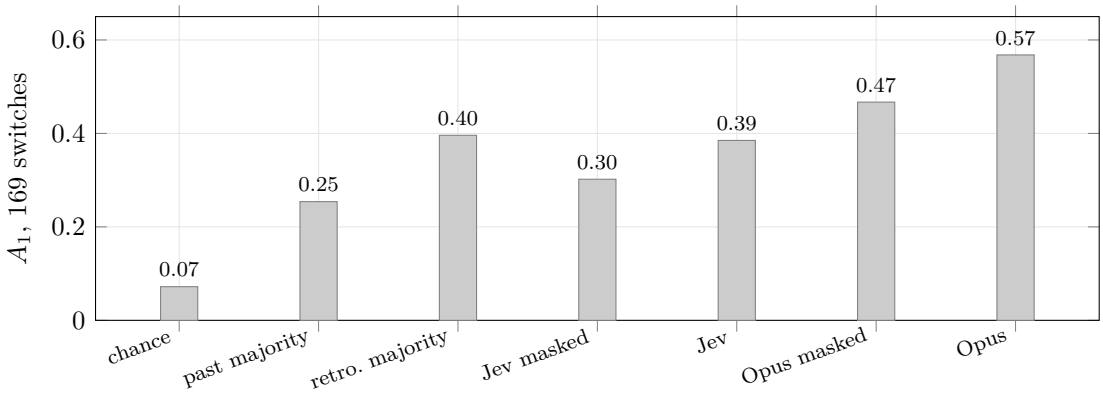


Figure 6: Decomposition of switch agreement. Masking every activity-derived field leaves a residual that depends on the reader: 0.30 for the classifier, 0.47 for Opus 5.5. The difference between masked and unmasked scores (0.08 and 0.10) is the contribution of activity fields once the sender has changed recipient.

switch is in progress, the limiting factor is the quality of the reading of the state rather than the quantity of state available.

4.4 From diagnostic to policy

The history conditions partition on the label and cannot be observed by a policy. The decomposition nevertheless identifies where a learned policy should be consulted: on switches, where past majority is incorrect in 71% of the decisions on which it votes (107 of 150), and on decisions without history, where it abstains. The design question is whether any signal observable before the decision predicts membership in those conditions.

The classifier's own confidence does not (mean top score 0.65 on continuations and 0.67 on switches in live shadow data). Peer arrival does, partially. Define the trigger "a candidate with no prior request participation is present and was active within T seconds". On known switches the recorded recipient lies in the triggered candidate set on 53.6% of activations at $T = 600$ (15 of 28); on all worker decisions the proportion is 27.0% (20 of 74), and on orchestrator decisions 37.5% (6 of 16). The trigger identifies a set containing the recipient rather than a recipient, and its precision conditional on a known switch is not its deployment precision.

Table 5 scores the policy "past majority unless the trigger fires, in which case select uniformly among triggered candidates". With history defined from the complete journal it exceeds past majority on switches (0.326 against 0.254) and on orchestrator switches (0.392 against 0.339) while preserving aggregate agreement. With history defined from the shorter ledger available to a live policy, most candidates appear novel, the trigger over-fires, and the policy loses approximately ten points to past majority on continuations. The availability of history is therefore part of the policy's specification. The threshold T was selected on this data; it is fixed at 600 seconds, and the live shadow track has scored the policy on all traffic since 13 October 2026. The prospective result constitutes the test of this section and is not yet available.

4.5 A contrast in regime

To verify that the policies rank stateless profile cards at all, and to exhibit the protocol on a dataset with a different history structure, we replayed the aware-c2 condition of DecisionBench [5] (691 delegations). The label is the orchestrator's recorded delegate rather than the benchmark's skill

Table 5: From diagnostic to policy. Agreement at rank one. The arrival policy selects by past majority unless a never-addressed candidate was active within 600 seconds, in which case it selects uniformly among such candidates.

Policy	Worker 555	Switch 169	Orch. switch 62	Evidence status
Past majority	.560	.254	.339	Recorded baseline
Jev	.578	.385	.371	Recorded policy
Opus 5.5	.775	.568	.500	Recorded policy, ∼70× cost
Arrival policy, journal history	.568	.326	.392	Exploratory replay
Arrival policy, ledger history	.511	.258	.284	History-availability sensitivity
Arrival policy, later traffic	pending	pending	pending	Prospective, frozen

annotation, so the exercise is a contrast rather than a replication. On the cards the classifier attains 0.175 at rank one against chance 0.096, within the benchmark's published fidelity range, and below an in-order majority of prior delegates (0.315): orchestrators concentrate on a few delegates in that setting as well. The history structure differs in kind. Of delegations with a previous delegate, 25.6% repeat it, against 66.3% in the mesh, and the classifier is flat across the same/switch partition (0.129 against 0.178). Figure 4 is not extended to DecisionBench because its conditional scores and history definition differ. The repeat share, reported with its denominator, indicates which regime a routing evaluation occupies and whether the mixture effect of equation 2 can alter its conclusions.

5 Related work

Model routing. RouterBench [7], LLMRouterBench [2] and TwinRouterBench [11] select which model answers a query or a trajectory step under a cost–quality trade-off. Workers are interchangeable by construction and have no history with the sender. TwinRouterBench's static and dynamic tracks, with router-visible prefixes and execution-verified labels, are the closest methodological relative of the present reconstruction; its execution grounding is precisely what recipient labels do not supply.

Multi-agent construction and delegation. MasRouter [12] and Uno-Orchestra [3] construct the agent system per task, selecting decomposition, roles, models and inference budget. DecisionBench [5] evaluates an orchestrator's choice of delegate from a pool of models with profile cards and reports that delegation fidelity can double while end-task quality remains unchanged. OrchBench [10] evaluates orchestration plans by simulation. CooperBench [1] and MultiAgentBench [13] evaluate agents that coordinate over a message bus or a fixed topology; neither conditions on sender history. The present work studies a distinct routing object, the selection of an existing recipient in a changing mesh. We do not claim that no earlier benchmark routes to stateful recipients; we claim that those above do not, and that the history structure under which the mixture effect operates is absent from them.

Claimed and attested state. The Provenance Paradox [9] distinguishes self-reported from observed capability and reports that self-report can route worse than chance. We adopt the terminology; because descriptions were not journaled, no comparison between the two is made here.

Repetition baselines and selective override. Session-based recommendation established that nearest-neighbour and repetition baselines are difficult to exceed and that evaluation must respect temporal order [8]; temporal link prediction draws the same conclusion for next-contact prediction [4]; selective prediction formalises the decision of when to answer [6]. The contribution here is not

these ideas but the measured failure mode and the reporting protocol for recipient selection among persistent agents.

6 Limitations

The study measures agreement with recorded recipients in one operator's mesh; it does not measure optimal recipients or downstream task utility. The registry is reconstructed from activity, descriptions are absent, and selection into the scorable set may favour observable peers; the 75 excluded recipients without prior evidence were most probably addressed at spawn, so the exclusion biases against the no-history findings. Almost all switches fall in one epoch (165 of 169 in May and June); eight occupied weeks are not shown to be independent, and joint grouping by week or sender yields five blocks, one containing 145 decisions. The lexical audit removes detected names and project mentions but not indirect identity cues. Thresholds and subgroup analyses are exploratory; model executions are single samples; masks retain activity-selected candidates; the frontier policies required a schema that prohibits abstention (Appendix C), so their scores are properties of the elicitation as well as of the model; hosted model identifiers carry no version, so execution dates are recorded in their place. The DecisionBench adapter differs in history semantics and labels. These boundaries delimit the case study; the protocol is applicable without them.

7 Conclusion

Among persistent agents most routing decisions repeat the preceding one, and an aggregate agreement score rewards that repetition. Conditioning on history separates one number into three that disagree: on the mesh studied, every learned policy is dominated by a one-line rule on continuations and dominates it on switches, so the pooled ranking is a property of the request mix. Masking activity attributes the continuation score of every model, including a frontier model, to recency encoded in the state; on switches a residual remains whose magnitude is set by the reader, from 0.30 for a small classifier to 0.47 for Opus 5.5. The design that follows is an inexpensive default with a learned override, and the open problem is the trigger rather than the ranker. The reconstruction procedure, the replay harness, the history-conditioned metric and a frozen prospective test are released so that subsequent policies for recipient selection among persistent agents can be evaluated on the decisions for which they are required.

Artifacts. [repository URL, data release terms, harness commands.]

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A Declared divergences from the live tracker

Each divergence is stated with its effect on the 169 worker switches. (1) *Ambiguous reply capture:* an acknowledgement with a message can be emitted without a captured reply when delivery fails; 15 historical requests are ambiguous and affect 20 scored switches. Excluding them (149 switches) moves past majority from 0.254 to 0.221, newest from 0.195 to 0.221, the classifier from 0.385 to 0.383 and the masked classifier from 0.302 to 0.309; no conclusion changes. (2) *Expiry:* the live tracker evicts lazily, the reconstruction expires at exactly 24 hours; re-conditioning at 24h10m and 48h changes no scored worker decision. (3) *Forgetting on reaping* is ignored; no scored decision changes. (4) *Self-addressed requests* are counted as inbound but do not update the sender's own last exchange, affecting the sender block of 74 of 1,001 snapshots; masked policies are unaffected. (5) Daemon restarts clear the live tracker and are not journaled, so the reconstruction contains fewer no-history decisions than a live system would.

B Lexical audit

For each decision the request text is matched, case-insensitively, against the recorded recipient's reconstructed metadata at that time. A *literal address* is a bounded full display name, exact peer identifier, or handle formed from the display name, the backend-stripped name or the parsed project name. A *project mention* is the complete parsed project field at non-alphanumeric boundaries, including occurrences inside paths. Literal addresses occur in 3 of 333 continuations, 1 of 169 switches and 0 of 53 no-history decisions; any reference (union of the two tests) occurs in 93, 58 and

20 respectively. The detector identifies references, not routing intent; shortened aliases, descriptive nicknames and indirect context are outside its scope.

C Elicitation of the frontier policies

Under a schema permitting an empty ranking, Opus 5.5 returned no ranking with zero confidence on approximately half of multi-candidate decisions; a stricter instruction did not alter this (5 of 10 on a fixed probe). The structured-output facility rejects a minimum array length above one, so the final schema is an object with one required numeric score per candidate key, ranked by score in descending order with ties broken in key order, plus a confidence field. On the same probe this schema yields complete rankings on every decision, sharply peaked scores, and 8 of 10 agreements. A narrow "abstained" outcome (complete answer, confidence exactly zero, all scores equal, at least two candidates) is scored as zero and does not trip the provider circuit breaker; it occurred once across the three models. The schema fails with a grammar-size error at 79 or more candidates (7 decisions per model). Haiku was executed with its default reasoning budget and produced 3.5 times the output tokens of the other two models.

D Cost

Classifier: 6,356 calls, 28.8M input tokens, approximately $1.21. Clef Flash: 555 calls, 2.2M input tokens, within a free allocation. Frontier policies: 555 calls each at 1.45M input tokens; output tokens 274k (Haiku), 77k (Sonnet), 79k (Opus); list cost $0.28, $3.67 and $7.37, plus $7.3 for the masked Opus execution. DecisionBench replay: approximately $0.65.